01、Autonomous Systems and Nonautonomous Systems.

这个模块学习系统的性质，讨论一些基本定义，

什么样的系统是稳定的？怎样探究系统的稳定性？什么样的系统是有意义的.

01.1 Autonomous Systems. 自控系统.

$\dot{x}(t) = f(x(t))$ $\quad x(0) = x_0$ $\longrightarrow$ dynamics 只与 x 有关，与 t 无关.

$f: D \to \mathbb{R}^n$. $D \subseteq \mathbb{R}^n$ open $\qquad (*)$

$f(0) = 0$ equilibrium point (EP) 在原点.

我们只考虑 EP 在原点的系统，因为所有系统，phase 都可以被转换到原点.

$\Longrightarrow$ 以更普遍的 Nonautonomous Systems 为例.

Suppose $\bar{y}$ is a solution of $\dot{y} = g(t, y)$

The change of variables $x(t) = y(t) - \bar{y}(t) \rightsquigarrow \dot{x}(t) = \dot{y}(t) - \dot{\bar{y}}(t) = g(t, x(t) + \bar{y}(t)) - \dot{\bar{y}}(t)$

$=: f(t, x(t))$

Since $\dot{\bar{y}}(t) = g(t, \bar{y}(t)) \Rightarrow f(t, 0) = 0, \ \forall t \geq 0$

"EP $x^*(t) = 0$, $\Rightarrow$ ① $f(t, 0) = 0$, ② $\dot{x}^*(t) = \dot{0} = 0 \Rightarrow$ EP x^* is also a solution."

01.1.1 Existance of solution (continuous)

$f(x)$ 连续，解一定存在.

$f: D \to \mathbb{R}^n$ is continuous at $x' \in D$ if

$\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x \in D$: $\|x - x'\| < \delta \Rightarrow \|f(x) - f(x')\| < \varepsilon$

if this satisfied for each $x \in D$, then f is continuous on D.

但并不保证解全局存在.

Theorem (Peano): if $f: D \to \mathbb{R}^n$ is continuous, then, for any $x_0 \in D$,

$\exists x: (-\varepsilon, \varepsilon) \to D$, $\varepsilon > 0$ for the system $(*)$ $\qquad$ compact set: 有界闭区间.

Further, given a compact set $U \subset D$, then

$\exists \alpha > 0$ s.t. $\forall x_0 \in U, \exists x: (-\alpha, \alpha) \to D$: $(*)$

这个东西一般是用不上，但你得了解.

01.1.2 Uniqueness of solutions (Lipschitz continuous)

常用：x^2 是 locally Lipschitz，$[-M, M]$，$M<\infty$

$|x|$ 是 globally Lipschitz.

在 $x=0$ 处连续但不可导.

f locally Lipschitz $\Longrightarrow$ unique solution.

Def: $f: D \to \mathbb{R}^n$ is

✳ • **locally Lipschitz** at $x' \in D$ if $\exists \delta>0$, $L>0$ $\|f(z)-f(y)\| \le L\|z-y\|$, $\forall z, y \in B(x', \delta)$

$B(x', \delta) = \{x \in \mathbb{R} \mid \|x-x'\|<\delta\}$

常取 $z=x'$，再让 $y \to x'$.

此时若 $\frac{\|f(z)-f(y)\|}{\|z-y\|} \to \infty$，则在 x' 不 Lipschitz.

$\Rightarrow$ $f(x)$ 有界. bounded by L

- locally Lipschitz on D, if at all $x' \in D$ Lipschitz.
- Lipschitz on $W \subset D$, if at all $x' \in W$ Lipschitz with a same L
- globally Lipschitz, if $\forall z, y \in \mathbb{R}^n$, it holds.

eg. $f(x) = \begin{bmatrix} x_1+x_2 \\ x_1 x_2 \end{bmatrix}$

Theorem (Picard-Lindelöf): if $f: D \to \mathbb{R}^n$ is locally Lipschit. then for each $x_0 \in D$.

$\exists!\ x: (-\varepsilon, \varepsilon) \to D$, $\varepsilon>0$ for the system.

$\|f_i(z)-f_i(y)\| \le L_i\|z-y\|$

$\|f(z)-f(y)\|^2 = \sum_{i=1}^n |f_i(z)-f_i(y)|^2 \le \sum_{i=1}^n L_i^2 \|z-y\|^2$

$\Longrightarrow \|f(z)-f(y)\| \le L\|z-y\|$

$L = \sqrt{\sum_{i=1}^n L_i^2}$

01.1.3 Lyapunov Stability $x^*=0$ is EP

Stability: $\forall \varepsilon>0$, $\exists \delta>0$: $\|x_0\|<\delta \Longrightarrow \|x(t)\|<\varepsilon$ $\forall t \ge 0$

当初值靠近原点，解不会远离原点.

Asymptotic stability: Stability + $\exists \delta>0$: $\|x_0\|<\delta \Longrightarrow x(t) \to 0$ as $t \to \infty$

当初值靠近原点，解向原点收敛.

✳ **Theorem (Lyapunov's direct method)**.

Let $f: D \to \mathbb{R}^n$ be continuous, if there exists a differentiable $V: D \to \mathbb{R}$

s.t. $V(x^*)=0$, $V(x)>0$ $\forall x \in D \setminus \{0\}$

$\dot{V}(x) = \underbrace{\frac{\partial V}{\partial x} f(x)}_{L_f V(x)} \le 0$ $\forall x \in D$

then x^* is stable

if further $\dot{V}(x)<0$, $\forall x \in D \setminus \{0\}$, then x^* is asymptotically stable.

if $\|x\| \to \infty \Rightarrow \|V(x)\| \to \infty$: globally ...

如果 $V(x)>0$ 或 $\dot{V}(x) \le 0$ / $\dot{V}(x)<0$ 只在一个 open ball $B_h \setminus \{0\}$ 中满足

$\Longrightarrow$ locally ...

LaSalle:

if $\dot{V}(x)=0$，这个条件带入 dynamics.

$\Longrightarrow x=0$ 那也可以 asy. stable

$\dot{V}(x) = -\frac{1}{2}x_1^2 \le 0$

$\dot{V}(x)=0 \Rightarrow x_1=0$

如果 $x_1=0$ 代入 dynamics 里

也得到 $x_2=0$，那也 asy. stable

01.2 Nonautonomous System 非自控系统.

$\dot{x}(t) = f(t, x(t)) \quad t \geq t_0 \geq 0 \quad x(t_0) = x_0$

$f: \mathbb{R} \times D \to \mathbb{R}^n \quad D \subset \mathbb{R}^n$ open.

$f(t,0) = 0$ EP.

f continuous $\rightsquigarrow$ Existance of Solutions

f locally Lipschitz on x $\rightsquigarrow$ Uniqueness of Solution

但稳定性不一样了

01.2.1 Stability

We need new stability definitions:

Stability: $x^* = 0$ is stable if

$\forall \varepsilon > 0, \forall t_0 \geq 0, \exists \delta > 0 : \|x_0\| < \delta \Rightarrow \|x(t)\| < \varepsilon \quad \forall t \geq t_0$

↑ can depend on t_0

Uniform Stability: $x^* = 0$, is uniformly stable if

$\forall \varepsilon > 0, \exists \delta > 0 : \forall t_0 \geq 0, \|x_0\| < \delta \Rightarrow \|x(t)\| < \varepsilon \quad \forall t \geq t_0.$

↑ independent of t_0

Asymptotic Stability: $x^* = 0$ is asy. stable if it is stable and.

$\forall t_0 \geq 0 \quad \exists c > 0 : \|x_0\| < c \Rightarrow x(t) \to 0 \text{ as } t \to \infty$

Uniform asymptotic Stability: $x^* = 0$ is uniformly asy. stable

if it is uniformly stable and

$\exists c > 0 : \forall t_0 \geq 0, \|x_0\| < c \Rightarrow x(t) \to 0$ uniformly as $t \to \infty$

Expotential Stability:

Definition: $x^* = 0$ is an expotentially stable EP (equilibrium point) of (*)

if $\exists c, \lambda, k > 0$ s.t. for all $t \geq t_0 \geq 0$ and all $\|x_0(t)\| < c$

$\|x(t)\| \leq k \|x_0(t)\| e^{-\lambda(t - t_0)}$

Remark: exp. stable $\Rightarrow$ uniform asy. stability.

Theorem (Lyapunov's direct method) $\dot{V}(t,x) = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t,x)$

Let $f: [0,\infty) \times D \to \mathbb{R}^n$ be continuous and let $x^* = 0 \in D$ be an EP.

if there is a differentiable $V: [0,\infty) \times D \to \mathbb{R}$ with

$$W_1(x) \le V(t,x) \le W_2(x) \qquad \forall t \ge 0,\ x \in D$$

$$\dot{V}(t,x) \le 0 \qquad \forall t \ge 0,\ x \in D$$

其实我们在选 $V(t,x)$ 时，往往不考虑 t，还是用 $V(t,x) = \frac{1}{2}x^2$，然后选 $W_1(x) = W_2(x) = V(x)$，再按需去推 $\dot{V}(x)$

with $W_1, W_2 : D \to \mathbb{R}$ continuous and positive definite, then $x^* = 0$ is uniformly stable.

If further $\dot{V}(t,x) \le -W_3(x) \quad \forall t \ge 0,\ \forall x \in D$

for some continuous, positive def. $W_3 : D \to \mathbb{R}$, then x^* is uniformly asy. stable.

if $D = \mathbb{R}^n$ and W_1 is radially unbounded, then x^* is globally uniformly asy. stable.

if there is a differentiable $V: [0,\infty) \times D \to \mathbb{R}$ and $k_1, k_2, k_3, a > 0$

s.t. (1) $k_1 \|x\|^a \le V(t,x) \le k_2 \|x\|^a$

(2) $\dot{V}(t,x) \le -k_3 \|x\|^a \qquad \forall t \ge 0,\ x \in D$

then $x^* = 0$ is exponentially stable. if $D = \mathbb{R}^n$, x^* is globally exp. stable.

01.2.2 Comparision function 与之后章节相关

Definition:

- A function $\alpha: [0, \delta) \to [0,\infty)$ is (of) "class $\mathcal{K}$" if it is continuous, strictly increasing, and $\alpha(0) = 0$ ↳ $\alpha \in \mathcal{K}$

  α is (of) "class $\mathcal{K}_\infty$" if $\delta = \infty$ and $\alpha(r) \to \infty$ for $r \to \infty$

- A function $\beta: [0,\delta) \times [0,\infty) \to [0,\infty)$ is (of) "class $\mathcal{KL}$" if it is continuous.

  $\beta(\cdot, s) \in \mathcal{K}$ for each fixed s, and for each fixed r, $\beta(r,\cdot)$ is strictly decreasing with $\lim_{s \to \infty} \beta(r,s) = 0$

$\mathcal{K}$ 是一个单增类，$\mathcal{K}_\infty$ 是一个单增类 $\to\infty$ 类.

$\mathcal{KL}$ 是一个对于第一项的单增，对于第二项的单减 类.

Properties of comparison functions:

- $\alpha \in \mathcal{K}$ on $[0,\delta) \Rightarrow \alpha^{-1}$ is defined on $[0, \alpha(\delta))$ and $\alpha^{-1} \in \mathcal{K}$
- $\alpha \in \mathcal{K}_\infty \Rightarrow \alpha^{-1} \in \mathcal{K}_\infty$
- $\alpha_1, \alpha_2 \in \mathcal{K} \Rightarrow \alpha_1 \circ \alpha_2 \in \mathcal{K}$ $\quad \alpha_1 \circ \alpha_2(r) = \alpha_1(\alpha_2(r))$ (same for $\mathcal{K}_\infty$)
- $\alpha_1, \alpha_2 \in \mathcal{K}$, $\beta \in \mathcal{KL} \Rightarrow \alpha_1(\beta(\alpha_2(r), s)) \in \mathcal{KL}$

我们可以利用这个重写我们的 Stability 定义.

The equilibrium $x^*=0$ of $\dot{x}(t)=f(t,x(t))$ is

- uniformly stable if and only if $\exists \alpha \in \mathcal{K}$ and $c>0$

  s.t. $\forall t \geq t_0$, $\forall \|x(t_0)\|<c$, $\|x(t)\| \leq \alpha(\|x(t_0)\|)$

- uniformly asy. stable iff $\exists \beta \in \mathcal{KL}$ and $c>0$

  s.t. $\forall t \geq t_0$, $\forall \|x(t_0)\|<c$, $\|x(t)\| \leq \beta(\|x(t_0)\|, t-t_0)$ (2.1)

- global uniformly asy. stable iff (2.1) holds for all $x_0 \in \mathbb{R}^n$

这是为了证明一些复杂的东西，我们这里不深入讨论.

$\alpha_1, \alpha_2 \in \mathcal{K}$

$\Rightarrow \alpha_1(\|x\|) < V(x) < \alpha_2(\|x\|)$

$V(x)$ pos. def.

if $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$

$V(x)$ radially unbounded

01.2.3 Converse Theorems

Theorem

Let $x^*=0$ be EP of $\dot{x}(t)=f(t,x(t))$ with

$f: [0,\infty) \times \mathbb{R}^n \to \mathbb{R}^n$ cont. differentiable and $\frac{\partial f}{\partial x}$ bounded in $\mathbb{R}^n$ uniformly in t

$\left(\left\| \frac{\partial f}{\partial x}(t,x) \right\| \leq L \right.$

$\left. \forall x \in \mathbb{R}^n \text{ and some } L>0 \right)$

if $x^*=0$ is globally exp. stable, there exists a cont. differentiable function V

$V: [0,\infty) \times \mathbb{R}^n \to \mathbb{R}^n$ and $c_1, c_2, c_3, c_4 > 0$ s.t.

$$c_1\|x\|^2 \leq V(t,x) \leq c_2\|x\|^2$$

$$\dot{V}(t,x) \leq -c_3\|x\|^2$$

$$\left\| \frac{\partial V}{\partial x} \right\| \leq c_4\|x\|$$

在这一模块中，只有 Lyapunov 是最有用的，其它东西都偏向证明.

但因为很多东西是理论基础，所以依然放在了这里.

我们只要会选出合适的 $V(x)$ / $V(t,x)$ 就已经很厉害了.

02. Systems with inputs

本章研究有输入系统的稳定性，并研究一些控制器的设计.

(外来输入 $u \to u(t)$, state 输入 $u = k(x)$)

02.1 input-to-state stability (ISS) 外部输入对系统的稳定.

$\dot{x} = f(x,u)$. $f: \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$

$u: t \to u(t)$ 是一个外部输入

也可以是 $\dot{x} = f(t,x,u)$

此时 $V = V(t,x)$

$\dot{V} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t,x,u)$

Definition: System (*) is input-to-state stable (ISS) if there exists $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for all $x_0 \in \mathbb{R}^n$

$$\|x(t)\| \le \beta(\|x_0\|, t) + \gamma\left(\sup_{\tau \in [0,t]} \|u(\tau)\|\right).$$

ISS $\xrightarrow{u=0}$ 0-GAS

不能有其它 EP.

02.1.1 Lyapunov-like conditions for ISS

Theorem: Suppose there exists a cont. differentiable $V: \mathbb{R}^n \to \mathbb{R}$ and $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$, $\alpha_3, \delta \in \mathcal{K}$ s.t.

$$\alpha_1(\|x\|) \le V(x) \le \alpha_2(\|x\|) \quad \forall x \in \mathbb{R}^n$$

$$\frac{\partial V}{\partial x} f(x,u) \le -\alpha_3(\|x\|) \quad \text{for all } \|x\| \ge \delta(\|u\|).$$

Then system (*) is input-to-state stable with $\gamma = \alpha_1^{-1} \circ \alpha_2 \circ \delta$

注意，这里有一个 θ trick. 常用

我们仍然取 $V(x) = \frac{1}{2}x^2$. 取 $\alpha_1(\|x\|) = \alpha_2(\|x\|) = V(\|x\|)$

计算 $\dot{V}(x)$ 时常出现这样的情况，例如.

$$\dot{V} = \underbrace{(\sin(t)-2)x^2}_{\le 0} + \underbrace{ux}_{\text{含 } u \text{ 的未定义项}}.$$

θ trick: $\theta \in (0,1)$

$$= \underbrace{(1-\theta)(\sin(t)-2)x^2}_{-\alpha_3(\|x\|)} + \underbrace{\theta(\sin(t)-2)x^2 + ux}_{\text{让这部分} \le 0, \text{得到 } \delta(\|u\|)}$$

$x=0$ 时它也=0，不考虑 $x=0$ 的情况.

$\theta(\sin(t)-2)x^2 + ux \le 0$

$\theta(2-\sin(t))x^2 \ge ux$

$\theta(2-\sin(t))|x|^2 \ge |u||x|$

$\theta(2-\sin(t))|x| \ge |u|$

$|x| \ge \frac{|u|}{\theta(2-\sin(t))}$

这是个逆推逻辑

- $\frac{\partial V}{\partial x} f(x,u) \le -\alpha_3(\|x\|)$ for all $\|x\| \ge \delta(\|u\|)$ is equivalent to:

$$\frac{\partial V}{\partial x} f(x,u) \le -\alpha_4(\|x\|) + \alpha_5(\|u\|) \text{ for some } \alpha_4, \alpha_5 \in \mathcal{K}$$

看起来有用但其实不常用.

而且可用 θ trick 代替.

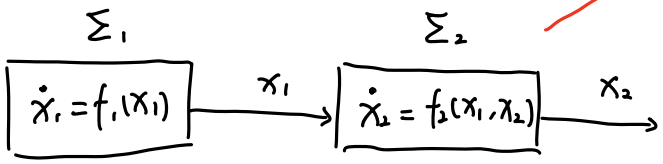
Theorem: Assume that f is globally Lipschitz, and $x^*=0$ is a globally exp. stable EP of $\dot{x}=f(x,0)$

很有趣，但几乎没用过.

Then system (*) is ISS.

02.1.2 cascaded System

An application of ISS: cascaded systems.



x_1 对于 Σ_2 来说是一个外部 input u.

如果 $x_1=0$ is GAS for Σ_1

Σ_2 ISS w.r.t x_1,

cascaded system $(x_1,x_2)=(0,0)$ GAS.

if • $x_1=0$ is a globally asy. stable EP for Σ_1

• Σ_2 is ISS w.r.t. x_1

Then $(x_1,x_2)=(0,0)$ is a globally asy. stable EP for the cascaded system.

02.2 Control Lyapunov function (CLF)

$$\dot{x}=f(x)+\sum_{i=1}^{m}g_i(x)u_i \qquad f:\mathbb{R}^n\to\mathbb{R}^n \quad g:\mathbb{R}^n\to\mathbb{R}^n$$
$$=f(x)+G(x)u \qquad G:\mathbb{R}^n\to\mathbb{R}^{n\times m}$$

Definition: A function $V:\mathbb{R}^n\to\mathbb{R}$ is a control Lyapunov function for (*) if it is differentiable, pos. def. radially unbounded, and.

$$\forall x\neq 0,\ \inf_u \nabla V(x)\cdot(f(x)+G(x)u)<0 \qquad (1)$$

这对 $\dot{x}=f(x,u)$ 这样的普遍系统也适用.

Theorem (Artstein): There exists $k:\mathbb{R}^n\to\mathbb{R}^m$ (state feedback) 在{0}处不一定连续.

continuous on $\mathbb{R}^n\setminus\{0\}$ such that the equilibrium $x^*=0$ is globally asy. stable under $u=k(x)$ if and only if there exists a control Lyapunov function.

如果存在 CLF，那么一定有 $u=k(x)$ 可以控制系统 GAS.

02.2.1 Find a CLF

$$\forall x \neq 0,\ \inf_u \nabla V(x)\cdot(f(x)+G(x)u) < 0 \qquad (1)$$

$$\forall x \neq 0:\ \frac{\partial}{\partial x}V(x)\,G(x)=0 \Longrightarrow L_f V(x) < 0 \qquad (2)$$

- $(2) \Leftrightarrow \forall x \neq 0,\ L_f V(x) \geq 0 \Rightarrow L_G V(x) \neq 0 \qquad (3)$

(1),(2),(3)是等价的，其实只要 $\nabla V(x)\cdot G(x)\neq 0$，它就是可控的，我们总能找到合适的 u.

但如果 $L_f V(x)<0$ 已经成立了，即使 $L_G V(x)=0$，它不可控，但也已经稳定，不需要控制了.

这就是(2),(3)的意义.

02.2.2 Continuity

$u=k(x)$，可能在 $\{0\}$ 处不连续，为找到一个连续控制器 u.

有{0}处连续但不平滑(可能破段)

Theorem: There exists a continuous $k:\mathbb{R}^n \to \mathbb{R}^m$, smooth on $\mathbb{R}^n\setminus\{0\}$ such that $x^*=0$ is globally asy. stable under $u=k(x)$ if and only if

(i) there exists a (smooth) CLF for (*)

(ii) $\forall \varepsilon>0\ \exists \delta>0$: for all $0<\|x\|<\delta$, there exists $u\in\mathbb{R}^m$ with $\|u\|<\varepsilon$ s.t. $L_f V(x)+L_G V(x)u<0$

这个就相当有难度了.

02.2.3 Sontag's formula to design controller u

"Sontag's formula": Fix $c\geq 0$, $a(x):=L_f V(x)$ and $b(x):=(L_G V(x))^T$

$$k(x)=\begin{cases} -c\,b(x)-\dfrac{a(x)+\sqrt{a(x)^2+(b(x)^T b(x))^2}}{b(x)^T b(x)}\,b(x), & b(x)\neq 0 \\ 0, & b(x)=0 \end{cases}$$

Proposition: Let $V:\mathbb{R}^n\to\mathbb{R}$ be a CLF and k as above. Then $x^*=0$ is globally asy. stable for $\dot{x}=f(x)+G(x)k(x)$

利用了CLF去design $u=k(x)$，一个非常好的通式.

02.3 Backstepping　利用 error coordinate，从上到下让系统一行一行稳定，得到 u.

这是一个过程复杂但并不难的 controller 设计方法，

没有新的概念和定义，我直接举例.

$$\begin{aligned}\dot{x}_1 &= x_1(x_1x_2 - x_1^2)\\ \dot{x}_2 &= \underbrace{x_2(x_2x_3 - x_2^2) - x_1^3}_{=\bar{u}_1}\\ \dot{x}_3 &= u\end{aligned}$$

$$\begin{aligned}\dot{x}_1 &= f_1(x_1) + g_1(x_1)x_2\\ \dot{x}_2 &= f_2(x_1,x_2) + g_2(x_1,x_2)x_3\\ &\vdots\\ \dot{x}_n &= f_n(x_1,\cdots x_n) + g_n(x_1,\cdots x_n)u\end{aligned} \quad (1)$$

① 首先，设计 $x_2 = \alpha_1(x_1)$，并找到 CLF $V_1(x_1)$ 使系统 $\dot{x}_1$ GAS.

$x_2 = 0 =: \alpha_1(x_1) \Rightarrow \dot{x}_1 = -x_1^3$，$V_1(x_1) = \frac{1}{2}x_1^2$，$\dot{V}_1(x_1) = -x_1^4 < 0$

选 $\alpha_1(x_1)$ 时注意，要满足 $(x_1, x_2-\alpha_1(x_1)) = (0,0) \Rightarrow (x_1,x_2) = (0,0)$

满足 $\alpha_1(0)=0$ 即可.

② 再写到 error coordinate 里

$(e_1,e_2) = (x_1, x_2-\alpha_1(x_1)) = (x_1, x_2)$

$\dot{e}_1 = e_1^2e_2 - e_1^3$

$\dot{e}_2 = \bar{u}_1$

再取 $V_2(e_1,e_2) = V_1(e_1) + \frac{1}{2}e_2^2 = \frac{1}{2}e_1^2 + \frac{1}{2}e_2^2$　也可取其它的.

$\dot{V}_2 = e_1^3e_2 - e_1^4 + \bar{u}_1e_2$

此时设计 $\bar{u}_1$，使 $\dot{V}_2 < 0$ (neg. def)，$\bar{u}_1 = -e_1^3 - e_2^3 \Rightarrow \dot{V}_2 = -e_1^4 - e_2^4 < 0$

消除不确定　添加 e_2 项.　这里取 $-e_2$ 其实也是对的，但为了方便且统一，取 $-e_2^3$ 让后续计算更简单.

$\Rightarrow \bar{u}_1 = -x_1^3 - x_2^3 \overset{!}{=} x_2^2x_3 - x_2^3 - x_1^3$

解出 $x_3 = 0 =: \alpha_2(x_1,x_2)$　要满足 $\alpha_2(0,0)=0$

③ 再一次写出 error coordinate.

$(e_1,e_2,e_3) = (x_1, x_2-\alpha_1(x_1), x_3-\alpha_2(x_1,x_2)) = (x_1, x_2, x_3)$

$\dot{e}_1 = e_1^2e_2 - e_1^3$

$\dot{e}_2 = e_2^2e_3 - e_2^3 - e_1^3$　这里要代入原本的，不能代入我们设计的 $\bar{u}_1$

$\dot{e}_3 = u$　这里如果是 ① $f(x_{1,2,3}) + g(x_{1,2,3})x_4$ 就设为 $\bar{u}_2$，解 $x_4 = \alpha_3(x_1,x_2,x_3)$ 重复

② $f(x_{1,2,3}) + g(x_{1,2,3})u$ 就设为 v，先设计 v，再代换成 u.

$V_3(e_1,e_2,e_3) = V_2(e_1,e_2) + \frac{1}{2}e_3^2$

$\dot{V}_3 = e_1^3e_2 - e_1^4 + e_2^3e_3 - e_2^4 - e_1^3e_2 + ue_3$

$= -e_1^4 - e_2^4 + e_2^3e_3 + ue_3$　设计 $u = -e_2^3 - e_3^3$　$\dot{V}_3 = -e_1^4 - e_2^4 - e_3^4 < 0$

$\Rightarrow u = -x_2^3 - x_3^3$　$(e_1,e_2,e_3) = 0 \Rightarrow (x_1,x_2,x_3) = 0$

$x=0$ is GAS.

这个方法对扰动很敏感，不如 Sontag's formula 更 robust.

注意：这里 $\alpha_1(x_1) = 0$，使换坐标系简化了.

实际上要复杂一点，

if $\alpha_1(x_1) = -x_1$，$e_2 = x_2 + x_1$

$\dot{e}_2 = \dot{x}_2 + \dot{x}_1$.

03. Systems with Inputs and Outputs

本章为系统引入输出 $y=h(x)$，提出有输出系统的稳定性的不同定义，
设计相对应的控制器 u。

03.1 Sliding-mode Control 设计一个 Sliding-mode surface，让解收敛于这里。

$$\dot{x} = f(x) + g(x)u + \delta(x) \quad \leftarrow \text{uncertainty}$$

$$f: \mathbb{R}^n \to \mathbb{R}^n \quad g: \mathbb{R}^n \to \mathbb{R}^n$$

$$y = h(x) = s(x) \qquad s: \mathbb{R}^n \to \mathbb{R}$$

- $\delta(x) = g(x)\delta(x)$ matched uncertainty
- $|\delta(x)| \le \beta(x)$ bounded

Sliding-mode surface. $S := \{x \mid s(x) = 0\}$ $s(0) = 0$.

① a) check $L_g s(x) \neq 0$, i.e. relative degree is 1, i.e. $\dot{y}$ 里含有 u.

b) remaining dynamics on S is AS.

② a) switching control law

$$u(x) = -\frac{1}{L_g s(x)} \left(L_f s(x) + \hat{u}(x) |L_g s(x)| \operatorname{sgn}(s(x)) \right)$$

one can also replace this by 1 if no uncertainty

b) high-gain property $\hat{u}(x) > \beta(x)$, $\forall x \in \mathbb{R}^n$

e.g. $\hat{u}(x) = \hat{u} + \beta(x)$, $\hat{u} > 0$

对于证明，因为我们已经检查过，dynamics remains on S 是 AS 的。
也就是说，一旦解收敛于 S，也会开始收敛到 origin。
所以只需证 $\longrightarrow s(x) \to 0$

① $\dot{y} = \dot{s}(x) = L_f s(x) + L_g s(x) u + \frac{\partial s}{\partial x} \cdot g(x)\delta(x)$ $\quad \rightarrow = L_g s(x) \cdot \delta(x)$

把 u 代进去 $= -\hat{u}(x)|L_g s(x)| \operatorname{sgn}(s(x)) + L_g s(x) \cdot \delta(x)$

$s(x) > 0$	< 0
$s(x) < 0$	> 0
$s(x) = 0$	$= 0$

得证

这个方法不够直观，但更适用于无 uncertainty 时，
它要取等，不能自由加绝对值。

② 选择 $V(x) = \frac{1}{2} s(x)^2$

$$\dot{V}(x) = s(x)\dot{s}(x)$$
$$= s(x)(L_f s(x) + L_g s(x) u + L_g s(x)\delta(x))$$
$$= s(x)(-\hat{u}(x)|L_g s(x)| \operatorname{sgn}(s(x)) + L_g s(x) \cdot \delta(x))$$
$$\le -|s(x)||L_g s(x)|(\hat{u}(x) - \delta(x))$$

$(\hat{u}(x) - \delta(x)) > 0$

$V \to 0$ in finite time, $x \to S$ in finite time

remaining dynamics on S is AS. independent of uncertainty

这个法更直观，因为不用取等可自由加绝对值号。

03.2 Dissipativity 耗散系统 稳定?

$\dot{x} = f(x,u)$ $\quad x(0) = x_0$ $\quad f: \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ $\quad u$: input

$y = h(x)$ $\quad h: \mathbb{R}^n \to \mathbb{R}^p$ $\quad y$: output.

- supply rate $s: \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$, $(u,y) \to s(u,y)$
- storage function $S: \mathbb{R}^n \to \mathbb{R}$, $x \to S(x)$ nonnegative. } 常用作 $V(x)$, $V(x) = S(x)$

Def: System is dissipative w.r.t. the supply rate s if there exists a storage function S

s.t. $\forall x_0 \in \mathbb{R}^n$, $\forall t \geq 0$, $\forall u: [0,t] \to \mathbb{R}^m$

$$S(x(t)) \leq S(x_0) + \int_0^t s(u(\tau), y(\tau))\, d\tau \quad (1)$$ dissipation inequality.

* $(1) \Longrightarrow \dot{S}(x) \leq s(u,y) = s(u, h(x))$ $\forall u, x$.

dissipation rate is define as

$$d(x,u) = s(u, h(x)) - \dot{S}(x) \geq 0$$ ← for dissipative systems.

03.2.1 Special Storage function

① "available storage" S_a:

$$S_a(x) := \sup_{\substack{u:[0,T]\to\mathbb{R}^m \\ T\geq 0,\ x(0)=x}} \left(-\int_0^T s(u(\tau), y(\tau))\, d\tau \right)$$

Theorem:

System (*) is dissipative w.r.t. supply rate s iff

$S_a(x) < \infty$ for all $x \in \mathbb{R}^n$. Moreover, if $S_a(x) < \infty$ for all $x \in \mathbb{R}^n$

then S_a is a storage function and $S_a \leq S$ for every other storage function S.

② "required supply" $S_r(x) = \inf_{\substack{u:[0,T]\to\mathbb{R}^m \\ T\geq 0,\ x(-T)=x^* \\ x(0)=x}} \int_{-T}^0 s(u(\tau), y(\tau))\, d\tau$.

Theorem:

Assum that each state $x \in \mathbb{R}^n$ is reachable from x^*.

if System (*) is dissipative. w.r.t supply rate s.

then for all storage functions $S: S(x) \leq S_r(x) + S(x^*) \quad \forall x \in \mathbb{R}^n$ (1)

Furthermore, $S_r(x) + S(x^*)$ is a storage function. $\Rightarrow S_r(x) > -\infty \quad \forall x \in \mathbb{R}^n$

03.2.2 Applications

① Dissipativity is widely used in control theory:

* • if System is dissipative with pos. def. storage function and there is continuous $k: \mathbb{R}^n \to \mathbb{R}^m$ s.t. $s(k(x), h(x)) < 0 \quad \forall x \neq 0$.

  Then origin is asy. stable under $u = k(x)$.

Why? Take S as (control) Lyapunov function:

$$\dot{S}(x) = \nabla S(x) \cdot f(x,u) \leq s(u, h(x)) \underset{u=k(x)}{=} s(k(x), h(x)) < 0 \quad \forall x \neq 0$$

② • L_2-gain via supply rate $s(u,y) = \frac{1}{2}\gamma^2 \|u\|^2 - \frac{1}{2}\|y\|^2$.

$\rightsquigarrow$ from DIE (dissipative inequality)

$$\frac{1}{2}\int_0^t \gamma^2 \|u(\tau)\|^2 - \|y(\tau)\| \, d\tau \geq S(x(t)) - S(x(0)) \geq -S(x(0))$$

$$\Rightarrow \int_0^t \|y(t)\|^2 \, d\tau \leq \gamma^2 \int_0^t \|u(\tau)\|^2 d\tau + 2S(x(0))$$

$\rightsquigarrow$ L_2-gain γ

03.3. Passivity　supply rate 为 $s(u,y)=u^Ty$ 的 dissipative 系统.

$$\dot{x}=f(x,u) \qquad x\in\mathbb{R}^n \quad u\in\mathbb{R}^m$$

$$y=h(x) \qquad y\in\mathbb{R}^m$$

Def: System is passive if it is dissipative w.r.t the supply rate $s(u,y)=u^Ty$

Def: System is zero-stable observable (ZSO) if $y(t)=0$ for all $t\geq 0 \Longrightarrow x(t)=0$ for all $t\geq 0$.
$(u=0)$

Theorem: Let system (*) be

(i) passive with differentiable storage function S with $S(0)=0$ and

(ii) ZSO.

Then, the feedback $u=-Py$, $P>0$, renders the origin asy. stable.

证明: $V(x)=S(x)$

$$\text{passive} \Longrightarrow \dot{S}(x)\leq u^Ty \overset{u=-Py}{=} -Py^Ty\leq 0. \qquad y=0 \overset{ZSO}{\Longrightarrow} x=0$$

from LaSalle. $\Longrightarrow$ origin asy. stable.

- input-feedforward passive if $\dot{S}(x)\leq u^Ty-u^T\varphi(u)$, $\varphi:\mathbb{R}^m\to\mathbb{R}^m$
- input-strictly passive if input-feedforward passive and
  $u^T\varphi(u)>0 \quad \forall u\neq 0$. ($\varphi(u)$常常等于 ku)
- output-feedback passive if $\dot{S}(x)\leq u^Ty-y^Ts(y)$, $s:\mathbb{R}^m\to\mathbb{R}^m$
- output-strictly passive if output-feedback passive and
  $y^Ts(y)>0 \quad \forall y\neq 0$ ($s(y)$常常等于 ky)

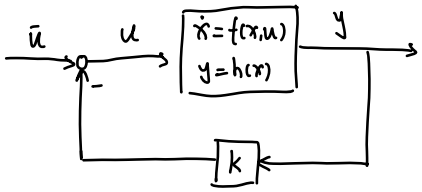
- "system that can be made passive by output feedback"

$\dot{S}(x) \le u^T y - \delta y^T y = \bar{u}^T y - (\delta + k) y^T y \le \bar{u}^T y$ if $k + \delta \ge 0$

$\delta < 0$ ... shortage of passivity

output-feedback $u = \bar{u} - ky$

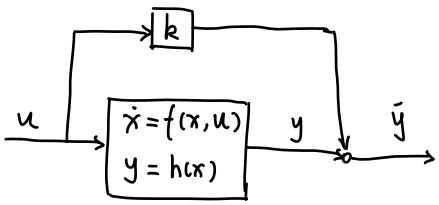
"new" reference input



- similar for "input-feedforward passivity"

system can be made passive by feeding forward the input.

$\bar{y} = y + ku$



$\dot{S}(x) \le u^T y - \gamma u^T u = u^T \bar{y} - (k + \gamma) u^T u \le u^T \bar{y}$

$\gamma < 0$ $\quad \bar{y} = y + ku \quad y = \bar{y} - ku$ $\quad$ if $k + \gamma > 0$

shortage of passivity δ 与 γ，在这里被定义为 < 0，前取负号.

但也可定义为 > 0，前取正号. 总之，它们是让系统差一点点 Passive 的因素.

excess of passivity 与 shortage 符号相反.



如果输出 y 满足 ZSO，我们可以不 feedback 与 feedforward

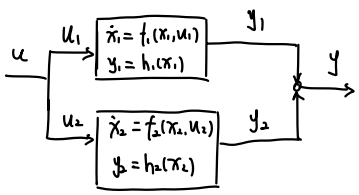
通过设计一个更严格的 u 来使系统在原点 AS.

$u = -Py$，$P > 0$，再加一些对 P 的限制.

这些限制可以通过令 $V(x) = S(x)$，$\dot{S}(x) \le -Py^T y +$ 关于 P 的项

代入 $u = -Py$ $\quad \le 0 \Rightarrow$ 得到 P 的限制.

(a) Parallel interconnections of passive systems are passive

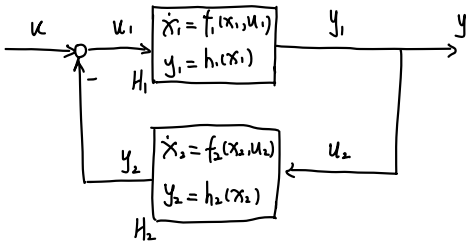


Take $S(x_1, x_2) = S_1(x_1) + S_2(x_2)$

Then $\dot{S}(x_1, x_2) \le u_1^T y_1 + u_2^T y_2 = u^T (y_1 + y_2) = u^T y$

- Does not work for series interconnections !

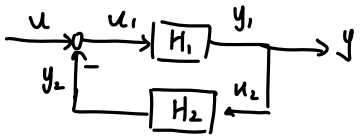
(b) Feedback interconnections of passive systems are passive.



Take $S(x_1, x_2) = S_1(x_1) + S_2(x_2)$

Then $\dot{S}(x_1, x_2) \le u_1^T y_1 + u_2^T y_2 = y^T (u_1 + y_2) = y^T u$

Theorem: ("Feedback Theorem for passive systems")



Consider the feedback interconnection (b) with $u \equiv 0$.

Assume that H_1 and H_2 are (i) ZSO

and (ii) dissipative w.r.t. the supply rate

$$(3)\quad s_i(u_i, y_i) = u_i^T y_i - \varrho_i y_i^T y_i - \nu_i u_i^T u_i, \quad i = 1, 2$$
$$\varrho_i, \nu_i \in \mathbb{R}$$

and differentiable storage functions S_i with $S_i(0) = 0$.

Then the origin $(x_1, x_2) = (0, 0)$ is asy. stable if

$$(4)\quad \boxed{\nu_1 + \varrho_2 > 0} \quad \text{and} \quad \boxed{\nu_2 + \varrho_1 > 0}$$

Proof: $S(x_1, x_2) = S_1(x_1) + S_2(x_2)$

$$\rightsquigarrow \dot{S}(x_1, x_2) \le u_1^T y_1 - \varrho_1 y_1^T y_1 - \nu_1 u_1^T u_1 + u_2^T y_2 - \varrho_2 y_2^T y_2 - \nu_2 u_2^T u_2$$

$$\underset{\substack{y_1 = u_2 \\ y_2 = -u_1}}{=} -\underbrace{(\varrho_1 + \nu_2)}_{>0} y_1^T y_1 - \underbrace{(\varrho_2 + \nu_1)}_{>0} y_2^T y_2 \le 0$$

$\rightsquigarrow$ can show that S is pos. def analogous to previous theorem $\rightsquigarrow$ stability.

for using LaSalle. $y_2 \equiv 0 \Rightarrow u_1 \equiv 0 \overset{\text{by ZSO of } H_1}{\Rightarrow} x_1 \equiv 0$

$y_1 \equiv 0 \Rightarrow u_2 \equiv 0 \Rightarrow x_2 \equiv 0$

$\Rightarrow$ asy. stable.

04. Input-Output Methods

定义 输入 $\longrightarrow$ 输出 的稳定, $\mathcal{L}_p$ 空间. 不再关心复杂的 state.

什么样的 输入 $\longrightarrow$ 输出 是好的信号?

04.1 Signal and System

04.1.1 $\mathcal{L}_p$-Space, $\mathcal{L}_p^e$-space and $\mathcal{L}_p$ Stability

- $\mathcal{L}_p$-spaces, $p \in [1, \infty)$

$$\mathcal{L}_p[0,\infty] = \{ \underline{e}: [0,\infty) \to \mathbb{R}^m \text{ measurable} \mid \int_0^\infty \|\underline{e}(t)\|^p dt < \infty \}$$

- $\mathcal{L}_p$-Norm

$$\|\underline{e}\|_{\mathcal{L}_p} = \sqrt[p]{\int_0^\infty \|\underline{e}(t)\|^p dt} \quad \text{for } \underline{e} \in \mathcal{L}_p.$$

i.e., (i) $\|\underline{e}\|_{\mathcal{L}_p} = 0 \iff \underline{e} = 0$, else $\|\underline{e}\|_{\mathcal{L}_p} > 0$.

(ii) for $\alpha \in \mathbb{R}$, $\|\alpha \underline{e}\|_{\mathcal{L}_p} = |\alpha| \cdot \|\underline{e}\|_{\mathcal{L}_p}$

(iii) for $\underline{e}_1, \underline{e}_2 \in \mathcal{L}_p$, $\|\underline{e}_1 + \underline{e}_2\|_{\mathcal{L}_p} \le \|\underline{e}_1\|_{\mathcal{L}_p} + \|\underline{e}_2\|_{\mathcal{L}_p}$.

"triangle inequality"

$$\|\underline{e}\|_{\mathcal{L}_\infty} = \inf \{ C \in \mathbb{R} \mid \|\underline{e}(t)\| \le C \quad a.e \}$$

$$* \left(= \sup_{t \ge 0} \|\underline{e}(t)\| \quad \text{for continuous } \underline{e} \right)$$

Specialcase $p=2$: $\underline{e}_1, \underline{e}_2 \in \mathcal{L}_2$, $\langle \underline{e}_1, \underline{e}_2 \rangle_{\mathcal{L}_2} = \int_0^\infty \underline{e}_1(t)^T \underline{e}_2(t)\, dt$

$\langle \underline{e}_1, \underline{e}_2 \rangle_{\mathcal{L}_2}$ 满足向量内积性质. ① $\langle \underline{e}_1, \underline{e}_2 \rangle_{\mathcal{L}_2} = \langle \underline{e}_2, \underline{e}_1 \rangle_{\mathcal{L}_2}$

② $\langle \alpha \underline{e}_1, \underline{e}_2 \rangle_{\mathcal{L}_2} = \alpha \langle \underline{e}_1, \underline{e}_2 \rangle_{\mathcal{L}_2}$ ③ $\langle \underline{e}_1 + \underline{e}_3, \underline{e}_2 \rangle_{\mathcal{L}_2} = \langle \underline{e}_1, \underline{e}_2 \rangle_{\mathcal{L}_2} + \langle \underline{e}_3, \underline{e}_2 \rangle_{\mathcal{L}_2}$

④ $\|\underline{e}\|_{\mathcal{L}_2}^2 = \langle \underline{e}, \underline{e} \rangle_{\mathcal{L}_2}$ ⑤ $|\langle \underline{e}_1, \underline{e}_2 \rangle_{\mathcal{L}_2}| \le \|\underline{e}_1\|_{\mathcal{L}_2} \cdot \|\underline{e}_2\|_{\mathcal{L}_2}$

$$\|\underline{e}\|_{\mathcal{L}_p}^p = \int_0^\infty |\underline{e}|^p dt = \int_0^\infty |\underline{e}|^{p-1} |\underline{e}|\, dt \le \int_0^\infty (\text{ess sup} |\underline{e}|)^{p-1} |\underline{e}| = \|\underline{e}\|_{\mathcal{L}_\infty}^{p-1} \|\underline{e}\|_{\mathcal{L}_1}$$

- Truncation operator

$$\mathbf{P}_T(t) = \begin{cases} \mathbf{P}(t) & 0 < t \leq T \\ 0 & t > T \end{cases}$$

- $\mathcal{L}_p^e$ - Space.

The extension on $\mathcal{L}_p^e$ of $\mathcal{L}_p$ is define as

$$\mathcal{L}_p^e = \{ \mathbf{P}: [0,\infty) \to \mathbb{R}^m \mid \forall T \geq 0, \mathbf{P}_T \in \mathcal{L}_p \} \supset \mathcal{L}_p.$$

$\mathcal{L}_p^e$ 不用绝对可积，但要保证在到达 ∞ 前绝对可积，不能在到达 ∞ 前趋向于 ∞

✳ - $\mathcal{L}_p$ - Stability. $u \in \mathcal{L}_p^e \longrightarrow y \in \mathcal{L}_p^e$ $\quad H: u \to y$.

Definition: • H is $\mathcal{L}_p$ stable if there exist $\alpha \in \mathcal{K}$, $\beta \geq 0$ s.t.

$$\| (Hu)_T \|_{\mathcal{L}_p} \leq \alpha(\| u_T \|_{\mathcal{L}_p}) + \beta \qquad \beta \dots \text{bias}.$$

for all $u \in \mathcal{L}_p^e$ and all $T \geq 0$

✳ • H is finite-gain $\mathcal{L}_p$ stable if there exist $\gamma, \beta \geq 0$ s.t.

$$\| (Hu)_T \|_{\mathcal{L}_p} \leq \gamma \| u_T \|_{\mathcal{L}_p} + \beta \quad (1)$$

for all $u \in \mathcal{L}_p^e$ and all $T \geq 0$

$|H(u)_T| \leq \gamma |u_T|$

γ 就是 $\mathcal{L}_p$ gain

Then, the $\mathcal{L}_p$-gain of H is $\gamma_p(H) = \inf \{ \gamma \mid \exists \beta \geq 0, \text{s.t. (1) holds} \}$

Definition: A map $H: \mathcal{L}_p^e \to \mathcal{L}_p^e$ is causal if

$$(Hu)_T = (Hu_T)_T \quad \forall u \in \mathcal{L}_p^e, T \geq 0 \iff u_T = v_T \Rightarrow Hu_T = Hv_T$$
$$\forall u, v \in \mathcal{L}_p^e, T \geq 0$$

04.1.2 Input-Output stability of state-space system

Theorem:

Consider $\dot{x} = f(x,u)$, $y = h(x,u)$

Suppose the system is ISS and there exist $\alpha_1, \alpha_2 \in \mathcal{K}$ and $\eta \geq 0$.

such that $\| h(x,u) \| \leq \alpha_1(\|x\|) + \alpha_2(\|u\|) + \eta$

Then, for each $x_0 \in \mathbb{R}^n$, the system is $\mathcal{L}_\infty$ - stable (bounded-input bounded-output)

✳ 一些简单应用.

① L_2-gain of LTI systems

Consider the SISO system $(*)\ \begin{cases} \dot{x} = Ax + Bu \\ y = Cx + D \end{cases}$.
where A is Hurwitz.

Lemma:

The L_2-gain of $(*)$ is $\gamma = \sup_{\omega \in \mathbb{R}} \sqrt{G(-i\omega)\, G(i\omega)}$, where $G(s) = C(sI - A)^{-1}B + D$, $s \in \mathbb{C}$

② L_2-gain of input-affine systems

Consider $\dot{x} = f(x) + g(x)u$, $y = h(x)$ $\quad (**)$ $\quad f: \mathbb{R} \to \mathbb{R}^n$, $g: \mathbb{R}^n \to \mathbb{R}^{n \times m}$, $h: \mathbb{R}^n \to \mathbb{R}^q$

Recall: System $(**)$ has L_2-gain $\le \gamma$ if it is dissipative w.r.t. the supply rate $s(u,y) = \frac{1}{2}\gamma^2 \|u\|^2 - \frac{1}{2}\|y\|^2$

$\rightsquigarrow$ with differentiable storage function

$$(1)\quad \frac{\partial S}{\partial x}\left[f(x) + g(x)u\right] - \frac{1}{2}\gamma^2\|u\|^2 + \frac{1}{2}\|h(x)\|^2 \le 0 \quad \forall x, u.$$

"worst-case" (maximizing) u for left-hand side

$$u^* = \frac{1}{\gamma^2} g(x)^T \left(\frac{\partial S}{\partial x}\right)^T$$

$\rightsquigarrow$ in to (1)

$$\frac{\partial S}{\partial x} f(x) + \frac{1}{2}\frac{1}{\gamma^2}\frac{\partial S}{\partial x} g(x) g(x)^T \left(\frac{\partial S}{\partial x}\right)^T + \frac{1}{2} h(x)^T h(x) \le 0 \quad \forall x \qquad (2)$$

Hamilton-Jacobi-inequality.

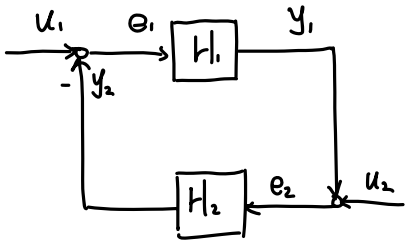
if there exists pos. semi def. $S: \mathbb{R}^n \to \mathbb{R}$ satisfying (2)

then system $(**)$ has L_2-gain $\le \gamma$

04.2 Feedback Theorems

这一章要习惯通过代换化简.

e.g. $e_1 = u_1 - y_2$, $e_2 = y_1 + u_2$



Theorem (Small-Gain Theorem)

Suppose that H_1 and H_2 are finite-gain $\mathcal{L}_p$ stable with gains γ_1 and γ_2

Then, the feedback interconnection is finite-gain $\mathcal{L}_p$ stable if $\gamma_1 \gamma_2 < 1$

Passive in Input-Output Method 对于 $\mathcal{L}_2$

Definition: $H: \mathcal{L}_2^e \to \mathcal{L}_2^e$ is passive if there exists $\beta \in \mathbb{R}$ s.t.

$$\forall u \in \mathcal{L}_2^e,\ T \geq 0: \quad \langle u_T, y_T \rangle_{\mathcal{L}_2} \geq -\beta$$

- output-strictly passive if there exist $\beta \in \mathbb{R}$ and $\varepsilon > 0$ s.t.

$$\forall u \in \mathcal{L}_2^e,\ T \geq 0: \quad \langle u_T, y_T \rangle_{\mathcal{L}_2} \geq \varepsilon \|y_T\|_{\mathcal{L}_2}^2 - \beta$$

(ε: "excess" of passivity)

Lemma similar for input-strictly passive

Let $H: \mathcal{L}_2^e \to \mathcal{L}_2^e$ be output-strictly passive with excess ε, Then H has $\mathcal{L}_2$-gain $\leq \frac{1}{\varepsilon}$

Theorem (Feedback Theorem for passive systems – I/O version)

suppose there exist $\varepsilon_i, \delta_i, \beta_i$ $i = 1.2$, such that

$$\langle (e_i)_T, H_i(e_i)_T \rangle_{\mathcal{L}_2} \geq \varepsilon_i \|H_i(e_i)_T\|_{\mathcal{L}_2}^2 + \delta_i \|(e_i)_T\|_{\mathcal{L}_2}^2 - \beta_i$$

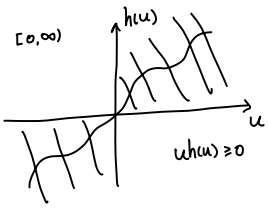
for all $T \geq 0$, $e_i \in \mathcal{L}_2^e$, $i = 1.2$ if

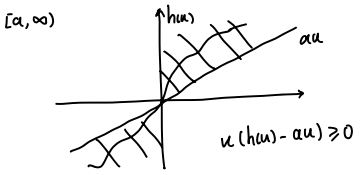
$$\varepsilon_1 + \delta_2 > 0 \quad \text{and} \quad \varepsilon_2 + \delta_1 > 0$$

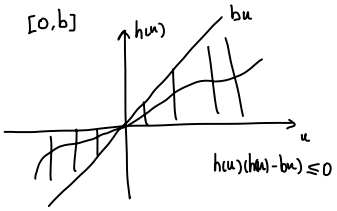
then the feedback interconnection has finite $\mathcal{L}_2$-gain from (u_1, u_2) to (y_1, y_2)

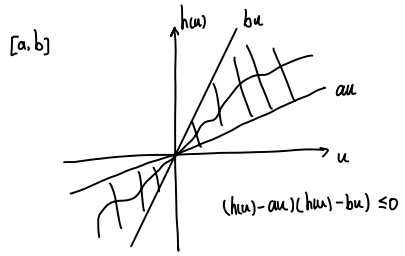
OS. 补充

Sector Nonlinearities: A scalar function $h: \mathbb{R} \to \mathbb{R}$, $u \to h(u)$ belongs to the sector "$h \in$ sector"









① $\int_0^t h(x)\dot{x}\,dt = \int_{x_0}^{x_{t+1}} h(\sigma)\,d\sigma$

$\dot{x}x = \frac{d}{dt}\frac{1}{2}x^2$

② $\|x^T y\| \le \frac{\|x\|^p}{p} + \frac{\|y\|^q}{q}$

$\forall x, y \in \mathbb{R}^n$, $\frac{1}{p} + \frac{1}{q} = 1$

$p, q > 1$

③ $\|x\|_1 \le \sqrt{n}\,\|x\|_2$ n is dimension.

④ $x = \begin{pmatrix} q \\ v \end{pmatrix}$ $H(x) = \frac{1}{2}v^T v + \frac{g}{m}P(q)$

$\dot{x} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \nabla H(x)$

$m\dot{v} = -g\nabla P(q)$

$S_r(x)$

consider $u: [-T, 0] \to \mathbb{R}^m$ transforming $x(-T) = \bar{x}$ to $x(0) = x$

dissipative: (from $-T$ to 0)

$S(x(0)) - S(x(-T)) = S(x) - S(\bar{x}) \le \inf \int_{-T}^{0} s(u(\tau), y(\tau))\,d\tau = S_r(x) \quad \forall x \in \mathbb{R}^n$

$\Rightarrow S(x) \le S(\bar{x}) + S_r(x) \Rightarrow S(\bar{x}) + S_r(x)$ nonnegative.

$S_r(x) = \inf \int_{-T}^{0} s(u(\tau), y(\tau))\,d\tau$

$\le \inf\left(\int_{-T}^{t} s(u(\tau), y(\tau))\,d\tau\right) + \int_{t}^{0} s(u(\tau), y(\tau))\,d\tau \quad \forall t \in [-T, 0]$

take $x(t) = \tilde{x}$

$= S_r(\tilde{x}) + \int_t^0 s(u(\tau), y(\tau))\,d\tau$

$\Rightarrow S_r(x) \le S_r(\tilde{x}) + \int_t^0 s(u(\tau), y(\tau))\,d\tau$.

$\Rightarrow S_r(x) + S(\bar{x}) \le S_r(\tilde{x}) + S(\bar{x}) + \int_t^0 s(u(\tau), y(\tau))\,d\tau$.

take $\bar{S}(x) = S_r(x) + S(\bar{x})$

$\Rightarrow \bar{S}(x) \le \bar{S}(\tilde{x}) + \int_t^0 s(u(\tau), y(\tau))\,d\tau$. satisfy DIE.

$S_a(x)$

Introduce "available storage" S_a:

$S_a(x) := \sup_{\substack{u:[0,T]\to\mathbb{R}^m \\ T\ge 0,\ x(0)=x}} \left(-\int_0^T s(u(\tau), y(\tau))\,d\tau\right)$

$S_a(x) = \sup\left(-\int_0^T s(u(\tau), y(\tau))\,d\tau\right)$

$\ge -\int_0^t s(u(\tau), y(\tau))\,d\tau + \sup\left(-\int_t^T s(u(\tau), y(\tau))\,d\tau\right) \quad \forall t \in [0, T]$

$\Rightarrow S_a(x) \ge -\int_0^t s(u(\tau), y(\tau))\,d\tau + S_a(x(t))$

$\Rightarrow S_a(x(t)) \le S_a(x(0)) + \int_0^t s(u(\tau), y(\tau))\,d\tau$

satisfy DIE.

判断 $\dot{V} < 0$ 技巧.

$V(x) = \frac{1}{2}(x_1^2 + x_2^2)$, $V(0) = 0$, V is pos. def.

$\dot{V} = \left\langle \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} -4x_1 + kx_2 \\ -2x_2 - x_1^2 + 2kx_1 \end{bmatrix} \right\rangle = -4x_1^2 + x_1^2x_2 - 2x_2^2 - x_1^2x_2 + 2kx_1x_2$

$= -4x_1^2 - 2x_2^2 + 2kx_1x_2$

asy. stable, $\Rightarrow \dot{V} < 0$

$\dot{V} = [x_1\ x_2] \begin{bmatrix} -4 & k \\ k & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\begin{bmatrix} -4 & k \\ k & -2 \end{bmatrix} = -\underbrace{\begin{bmatrix} 4 & -k \\ -k & 2 \end{bmatrix}}_{\text{pos. def.}}$

$\begin{cases} 4 > 0 \\ \det\begin{vmatrix} 4 & -k \\ -k & 2 \end{vmatrix} = 8 - k^2 > 0 \Rightarrow k \in (-2\sqrt{2}, 2\sqrt{2}) \end{cases}$

结束在2.0. 卷子不简单. 但特别难的题也不多.

题量很大, 12道题. 有一半以上题是不太难的.

40分左右满分. 15分及格.